

EDUCATION

IIT Roorkee — BTech, Mechanical Engg (2021–25)    — JEE Advanced AIR 2817    — JEE Main AIR 1429

PUBLICATIONS

Spectral Band Attention NetworkJan 2025 – July 2025

Computer Vision — [Code](#)    [Paper](#)

- Classification of wheat seeds into 96 varieties; finetuned DenseNet-121, ResNet-50, and GoogleNet for RGB image classification (max 78% accuracy) and developed DenseNet-121 inspired architecture for Hyperspectral data.
- Applied **Spectral Band Attention** Module (SBAM) for band selection, ranking bands by their contribution to classifier accuracy, achieving 87% accuracy on Hyperspectral images.
- Employed Regression-based Ensemble** (SVM) to combine model predictions, enhancing overall accuracy to 95%.

EXPERIENCE

Avathon — Software DeveloperJuly 2025 – Present

avathon.com

- Developed even-driven distributed task infrastructure using **RabbitMQ** with exponential backoff and real-time monitoring, increasing system throughput and eliminating blocking operations across microservices.
- Enhanced **Neon** (proprietary Scala graph DB) by implementing **IQL (Integrated Query Language)** support & semantic search, enabling complex relational querying & data retrieval for supply chain solutions.
- Support for access-based control for namespaces and graph data in Neon, ensuring secure **multi-tenant** isolation and granular permissions across the distributed datastore.
- Full backend support using **FastAPI** for multi-hop traversal along the graph nodes, dynamic IQL generation, and various CRUD usecases in Neon.
- Developed an AI-driven supply chain platform hosted on **GCP** using **Docker** and FastAPI, featuring a **LangChain**-based Demand Planning Agent for advanced anomaly detection, an LLM document parser for automated data extraction, and a web scraping service providing real-time shipment visibility.

DeepLogic AI — AI Engineer InternJuly 2024 – Dec 2024

Certificate

- Built a **RAG** pipeline for enterprise search, managing email and document embeddings in a Postgres vector-store on **AWS** for high-performance retrieval.
- Designed normalized schemas & CRUD operations for metadata-filtered searches across millions of docs.
- Integrated Retriever, Response Generator, and Re-ranker with caching strategies, improving chatbot reliability.

PROJECTS

NeurIPS 2025 Pokemon AI CompetitionSep 2025 – Nov 2025

Game AI — [Leaderboard](#)    [Imitation Learning](#), [Offline RL](#)

- Secured **Global Rank 3** in Track-1 with **ELO rating of 1793**, superior generalization in unseen scenarios.
- Trained an **Actor-Critic policy** using **Offline RL** on replay datasets, with exploratory **Behavior Cloning**.
- Further developed a **policy ensemble** to mitigate high action-entropy, dynamically selecting confident agents based on **TD error** to handle less explored scenarios during behaviour cloning.

Enriched Bots — Clever ChatJune 2024 – July 2024

Generative AI — [Demo](#)    [Python](#), [Django](#), [Langchain](#), [Hugging Face](#)

- Enhanced text-based interactions with multimedia using a novel graph-based architecture.
- Decomposed queries into acyclic graphs to store text and media metadata within nodes.
- Generated responses via topological traversal to ensure logical sequencing of enriched output.

AI Agent 007: Tooling up for Success (Inter-IIT Techfest 2023)Dec 2023

Generative AI — [Code](#)    [Python](#), [Langchain](#), [GPT-4](#), [Hugging Face](#)

- Built a query-aware agent capable of allocating and reviewing tool outputs.
- Focused on creating autonomous tools for efficient parameter extraction and downstream function calls.
- Implemented a self-reflective ReAct style agent; curated dataset using given tool descriptions.

Sentinel-2 Field DelineationJuly 2024 – August 2024

Computer Vision — [Code](#)    [PyTorch](#), [OpenCV](#), [Segmentation Models](#)

- Engineered a **field delineation** pipeline using high-resolution **hyperspectral imagery** from Solafune.
- Fine-tuned **UNet++**, **FPN**, & **Mask-RCNN**; extracted annotation polygons from patches via **OpenCV**.
- Designed an **ensemble model** by stacking prediction masks, achieving a segmentation **IoU of 0.96**.

ACHIEVEMENTS

- Global Rank 3 in NeurIPS 2025 Pokemon AI Competition (Track-1), competing as Team [\[srsk-1729\]](#)
- Runner Up in AI Agent 007: Tooling up for Success (Inter-IIT Techfest 2023) sponsored by DevRev

TECHNICAL SKILLS

AI & ML: PyTorch, TensorFlow, LangChain, Spark, SQL, Hugging Face, RAG, PEFT/LoRA, OpenCV  
Dev & Ops: Docker, AWS, Linux, FastAPI, Django, MLOps, SageMaker, Git, AsyncIO

CERTIFICATIONS

- [Machine Learning Certificate](#)
- [AWS Cloud Practitioner Certificate](#)
- [Scala Programming Certificate](#)